

1 Summary

Model (Run-name): model1 (test)

Iteration: 15

TAC: 154724.1846684643

Took: 1.6423521041870117s

Total Time: 20.201141357421875s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

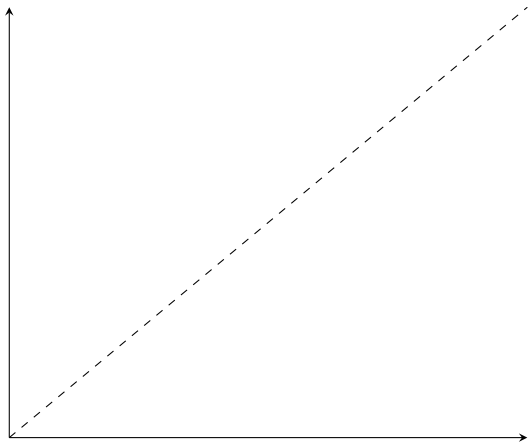
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

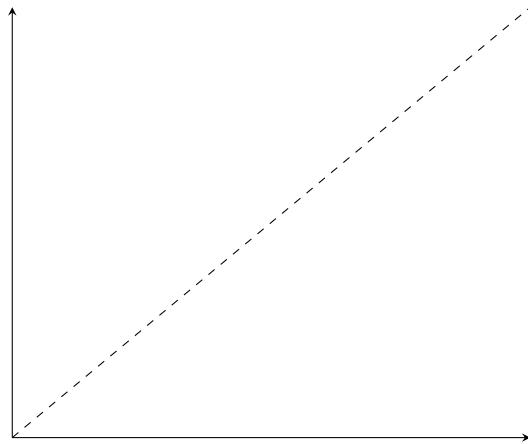
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$RecLMTD_{1,1,1}$	0.052206893965638275	$b_{1,1,1}^{H,out}$	899.2976047638144
$z_{1,2,2}$	1.0	$RecLMTD_{1,2,2}$	0.01770037069507927	$b_{1,2,2}^{H,out}$	3869.4504612598157
$z_{2,1,2}$	1.0	$RecLMTD_{2,1,2}$	0.02904476541544918	$b_{2,1,2}^{H,out}$	9370.824308110272
$z_{cu,1}$	1.0	$RecLMTD_{cu,1}$	0.014605996451365417	$b_{1,1,1}^{C,in}$	8543.444606394036
$z_{cu,2}$	1.0	$RecLMTD_{cu,2}$	0.009579189587726342	$b_{1,2,2}^{C,in}$	4550.0
$z_{hu,1}$	1.0	$RecLMTD_{hu,1}$	0.022546187384804434	$b_{2,1,2}^{C,in}$	4192.165337294863
$A_{1,1,1}^\beta$	70.50425945122109	$q_{1,1,1}$	680.5495387401838	$b_{1,1,1}^{C,out}$	9223.99414513422
$A_{1,2,2}^\beta$	69.0314457108091	$q_{1,2,2}$	1950.0	$b_{1,2,2}^{C,out}$	6500.0
$A_{2,1,2}^\beta$	140.7008934199032	$q_{2,1,2}$	2429.175691889724	$b_{2,1,2}^{C,out}$	6621.341029184586
$A_{cu,1}^\beta$	4.849719301830286	$q_{cu,1}$	169.4504612598157		
$A_{cu,2}^\beta$	37.682669717644174	$q_{cu,2}$	1970.8243081102764		
$A_{hu,1}^\beta$	13.218222920629291	$q_{hu,1}$	490.2747693700919		
$A_{1,1,1}$	70.50425945122109	$t_{1,1}^{(H)}$	650.0		
$A_{1,2,2}$	69.0314457108091	$t_{1,2}^{(H)}$	581.9450461259815		
$A_{2,1,2}$	140.7008934199032	$t_{1,3}^{(H)}$	386.94504612598155		
$A_{cu,1}$	4.849719301830286	$t_{2,1}^{(H)}$	590.0		
$A_{cu,2}$	37.682669717644174	$t_{2,2}^{(H)}$	590.0		
$A_{hu,1}$	13.218222920629291	$t_{2,3}^{(H)}$	468.5412154055138		
$dt_{1,1,1}$	32.68498462467277	$t_{1,1}^{(C)}$	617.3150153753272		
$dt_{1,1,2}$	10.0	$t_{1,2}^{(C)}$	571.9450461259815		
$dt_{1,2,2}$	81.94504612598155	$t_{1,3}^{(C)}$	410.0		
$dt_{1,2,3}$	36.94504612598158	$t_{2,1}^{(C)}$	500.0		
$dt_{2,1,2}$	18.05495387401845	$t_{2,2}^{(C)}$	500.0		
$dt_{2,1,3}$	58.54121540551381	$t_{2,3}^{(C)}$	350.0		
$dt_{cu,1}$	66.94504612598158	$t_{1,1,1}^H$	370.0		
$dt_{cu,2}$	148.5412154055138	$t_{1,2,2}^H$	386.94504612598155		
$dt_{hu,1}$	62.68498462467279	$t_{2,1,2}^H$	468.5412154055136		
$f_{1,1,1}^H$	2.430534066929228	$t_{1,1,1}^C$	617.507387405553		
$f_{1,2,2}^H$	10.0	$t_{1,2,2}^C$	500.0		
$f_{2,1,2}^H$	20.0	$t_{2,1,2}^C$	647.576992685471		
$f_{1,1,1}^C$	14.93746363729943	$b_{1,1,1}^{H,in}$	1579.8471435039983		
$f_{1,2,2}^C$	13.0	$b_{1,2,2}^{H,in}$	5819.450461259816		
$f_{2,1,2}^C$	10.224793505597226	$b_{2,1,2}^{H,in}$	11800.0		

3 *RecLMTD* Tangent Points

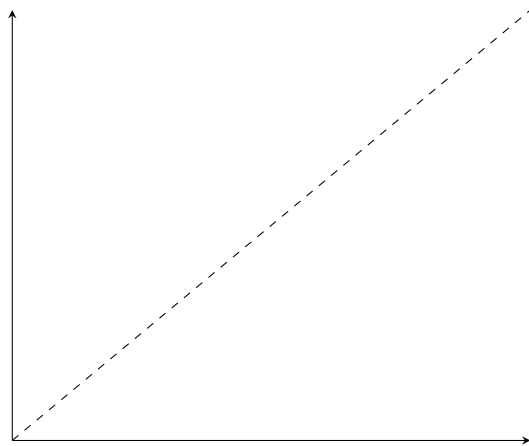
- $(1, 1, 1)$
 - Value: 0.052207
 - Error: 0.000000798182107
 - Relative error: 0.000015288592055
- $(1, 2, 2)$
 - Value: 0.017700
 - Error: 0.000002235630727
 - Relative error: 0.000126288224790
- $(2, 1, 2)$
 - Value: 0.029045
 - Error: 0.000009805160837
 - Relative error: 0.000337473954796
- $cu, 1$
 - Value: 0.014606
 - Error: 0.000000825166030
 - Relative error: 0.000056491826355
- $cu, 2$
 - Value: 0.009579
 - Error: 0.000000076770945
 - Relative error: 0.000008014282364
- $hu, 1$
 - Value: 0.022546
 - Error: 0.000000084736441
 - Relative error: 0.000003758334881



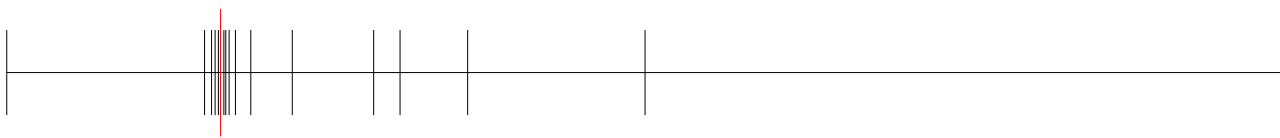
(a) *RecLMTD* - (1, 1, 1)



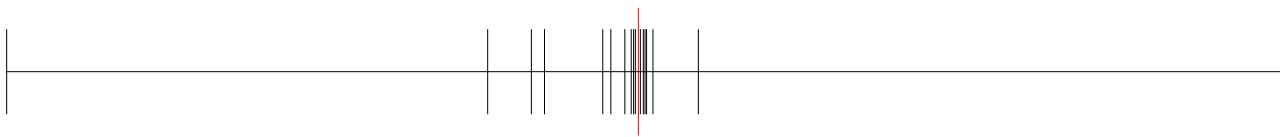
(b) *RecLMTD* - (1, 2, 2)



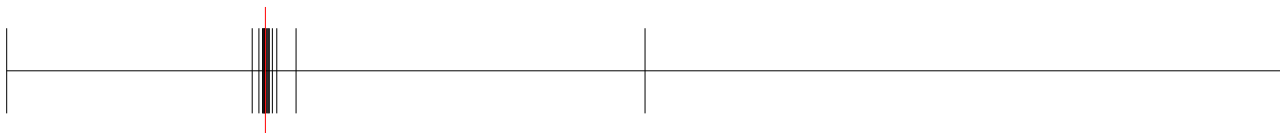
(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - *cu*, 1



(b) *RecLMTD* - *cu*, 2



(a) *RecLMTD* - *hu*, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 1579.8471435039983
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5819.450461259816
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 11800.0
 - Absolute error: 0.0
 - Relative error: 0.0

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 899.2976047638144
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 3869.4504612598157
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 9370.824308110272
 - Absolute error: 0.0
 - Relative error: 0.0

4.3 $b^{C,\text{in}}$

- (1, 1, 1)
 - Value: 8543.444606394036
 - Absolute error: 0.03627735364170803
 - Relative error: 4.246239000235489e-06
- (1, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
 - Value: 4192.165337294863
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 9223.99414513422
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 6621.341029184586
 - Absolute error: 9.094947017729282e-13
 - Relative error: -1.3735808165810965e-16



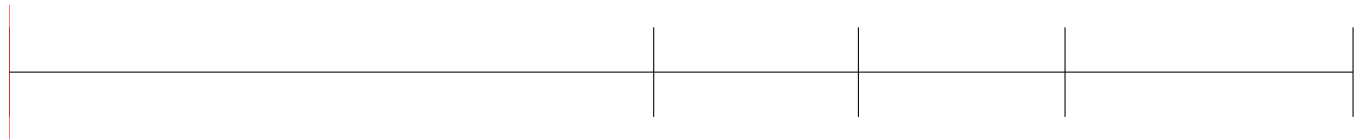
(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



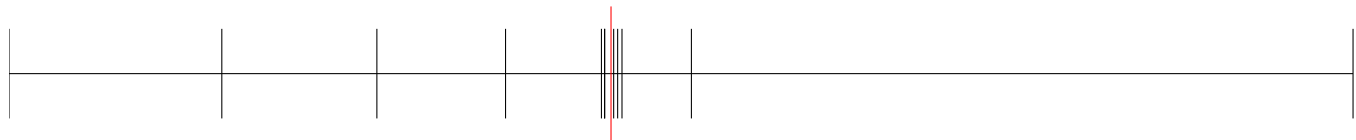
(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



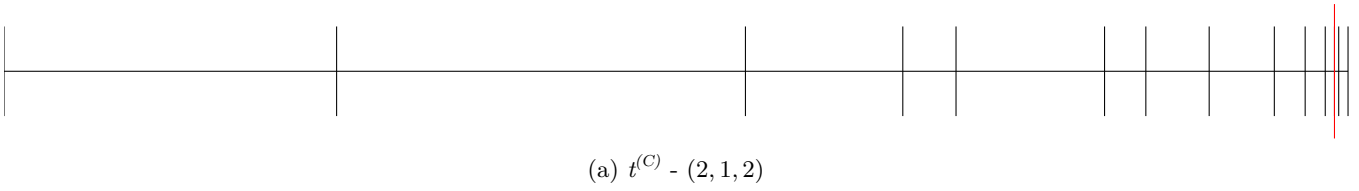
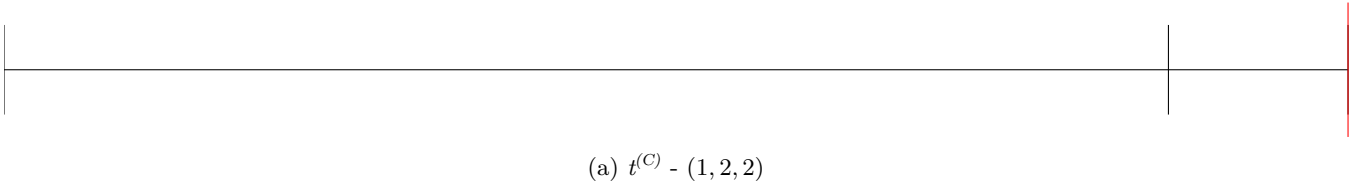
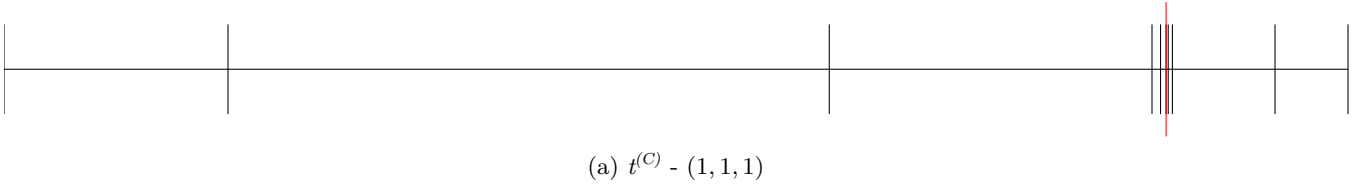
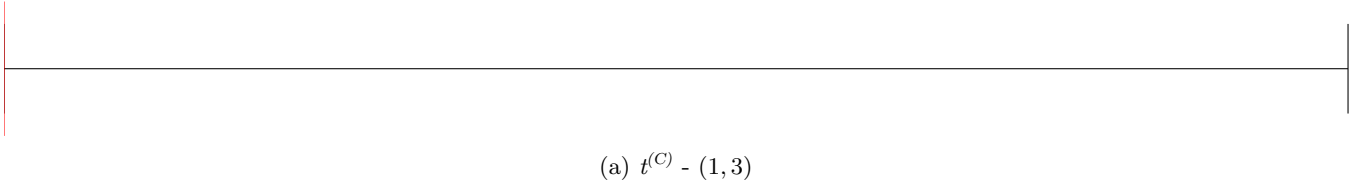
(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$



5 Area Breakpoints

5.1 A

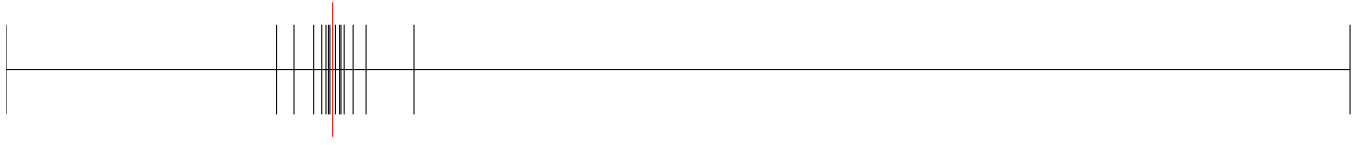
- $(1, 1, 1)$
 - Value: 70.504259
 - Error: 0.554495763524542
 - Relative error: 0.007803341922453
- $(1, 2, 2)$
 - Value: 69.031446
 - Error: 0.0000000000000057
 - Relative error: 0.000000000000001
- $(2, 1, 2)$
 - Value: 140.700893
 - Error: 0.408782827793772
 - Relative error: 0.002896915637991

5.2 A_{CU}

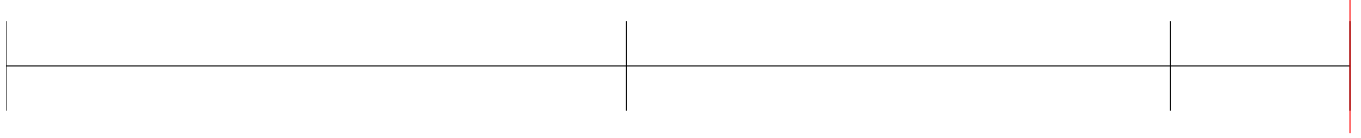
- $cu, 1$
 - Value: 4.849719
 - Error: 0.100266369855916
 - Relative error: 0.020255890927006
- $cu, 2$
 - Value: 37.682670
 - Error: 0.075129665331687
 - Relative error: 0.001989778709550

5.3 A_{HU}

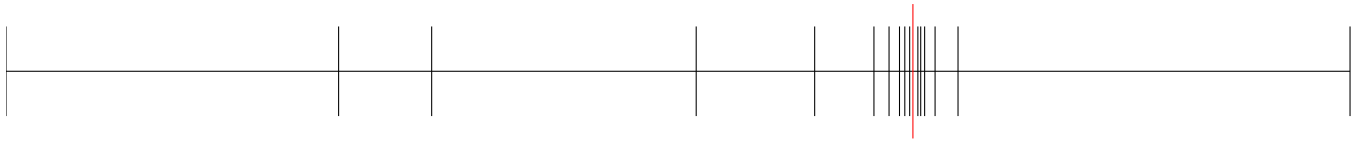
- $hu, 1$
 - Value: 13.218223
 - Error: 0.046369263682552
 - Relative error: 0.003495717247711



(a) $A^\beta - (1, 1, 1)$



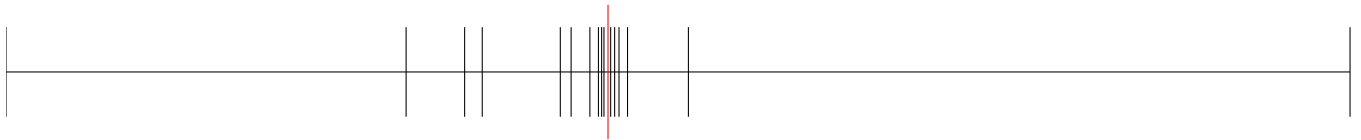
(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$

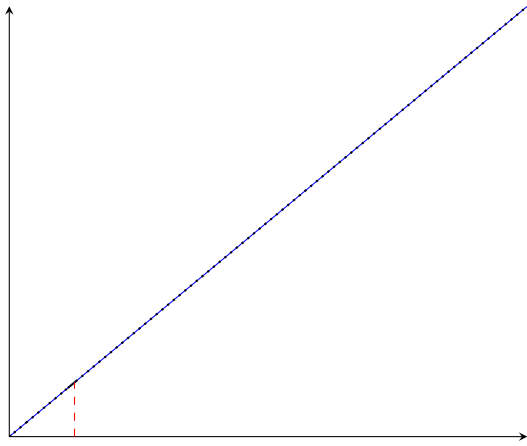


(a) $A^\beta - hu, 1$

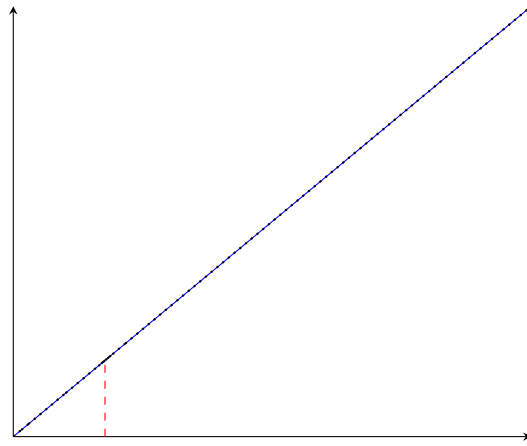
6 A^β Breakpoints

- (1, 1, 1)
 - q : 70.504259
 - $\hat{q}^\beta$: 70.504259
 - q^β : 70.504259
 - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 67.27362048222933, 69.14107665413653, 69.42777867315647, 69.89902342903967, 70.22290217272871, **70.30181758862284**, **71.22933662920877**, 71.6637831223933, 71.88123217857378, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1, 2, 2)
 - q : 69.031446
 - $\hat{q}^\beta$: 69.031446
 - q^β : 69.031446
 - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 66.17325582523148, 67.27673099173637, **67.91344539650363**, **69.34502840170047**, 70.94081287188642, 72.25348351595485, 74.6496383881616, 84.1468297226002, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2, 1, 2)
 - q : 140.700893
 - $\hat{q}^\beta$: 140.700893
 - q^β : 140.700893
 - Breakpoints: [0, 9.898970403313276, 28.33333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 118.88765652962957, 128.52021081477858, 131.69760457629607, **132.12228195956155**, **144.70843093515643**, 147.07009017600456, 147.48374761404892, 149.84689443310864, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 4.849719
 - $\hat{q}^\beta$: 4.849719
 - q^β : 4.849719
 - Breakpoints: [0, 2.5450346292088537, 2.9819181007907165, 3.1809419553840232, 3.9948823402911255, 4.269280092716329, **4.432280503220532**, **4.983938626310074**, 4.993068227518925, 5.0418124100090695, 7.702824927089129, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 2$
 - q : 37.682670

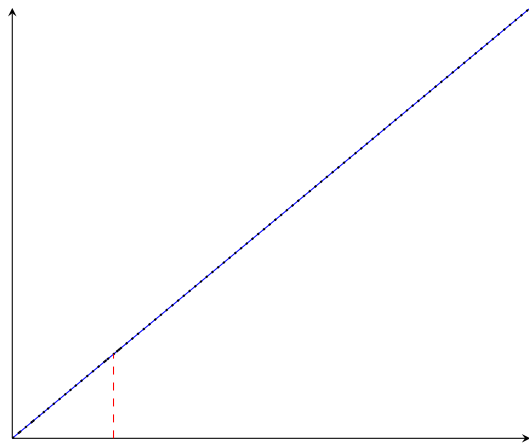
- $\hat{q}^\beta$: 37.682670
- q^β : 37.682670
- Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717, 36.670233878040335, 37.21616748032837, 37.443498367229196, 37.47875108132638, **37.554741031910254**, **37.9454585315849**, 38.10872593446926, 38.21652191748154, 38.88483572219945, 285.4001551947793]
- Error: 0.0000000000000000
- Relative error: 0.0000000000000000
- $hu, 1$
 - q : 13.218223
 - $\hat{q}^\beta$: 13.218223
 - q^β : 13.218223
 - Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, 12.917705220652834, 13.104034300095861, 13.161020122653593, **13.192252946941991**, **13.22717920580809**, 13.244985069124171, 13.30871858348944, 13.346967780023082, 13.439500729482463, 13.468285806082731, 14.09178614751222, 237.3002543523117]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



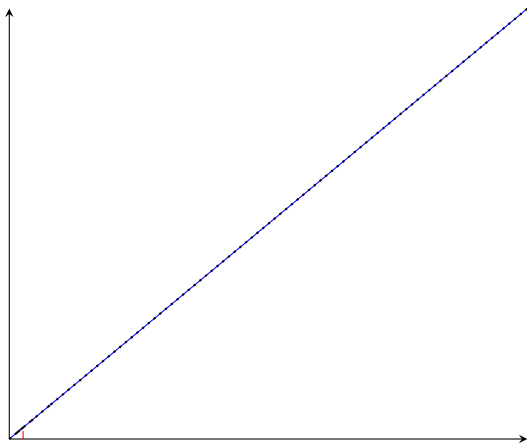
(a) $q^\beta - (1, 1, 1)$



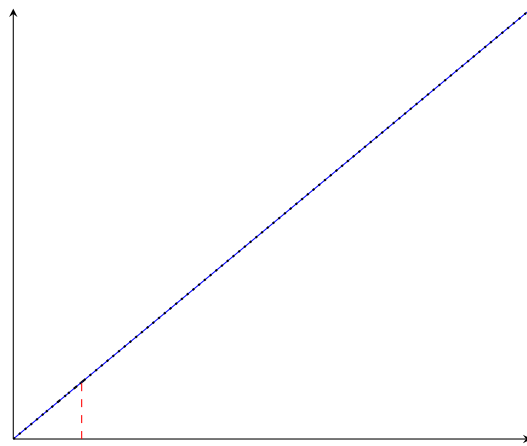
(b) $q^\beta - (1, 2, 2)$



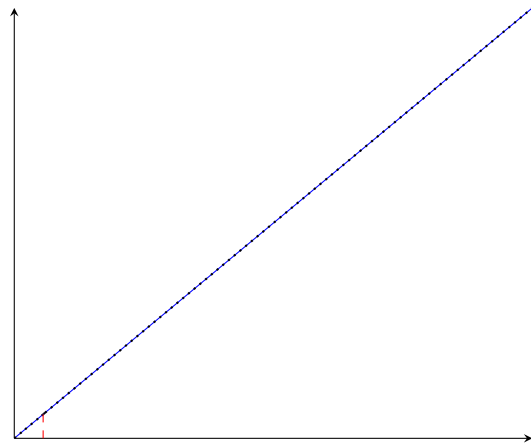
(c) $q^\beta - (2, 1, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	-0.0	$f_{2,2,1}^C$	0.0	$b_{2,2,2}^{H,out}$	0.0
$z_{1,2,1}$	-0.0	$f_{2,2,2}^C$	0.0	$b_{1,1,2}^{C,in}$	1957.8346627051371
$z_{2,1,1}$	0.0	$RecLMTD_{1,1,2}$	0.1	$b_{1,2,1}^{C,in}$	6500.0
$z_{2,2,1}$	-0.0	$RecLMTD_{1,2,1}$	0.008076065637336493	$b_{2,1,1}^{C,in}$	35.73108549568924
$z_{2,2,2}$	-0.0	$RecLMTD_{2,1,1}$	0.02916666666666668	$b_{2,2,1}^{C,in}$	0.0
$z_{hu,2}$	-0.0	$RecLMTD_{2,2,1}$	0.020111728838038165	$b_{2,2,2}^{C,in}$	0.0
$A_{1,1,2}^\beta$	0.0	$RecLMTD_{2,2,2}$	0.00434027777777778	$b_{1,1,2}^{C,out}$	1957.8346627051387
$A_{1,2,1}^\beta$	0.0	$RecLMTD_{hu,2}$	0.005549329826813255	$b_{1,2,1}^{C,out}$	6500.0
$A_{2,1,1}^\beta$	0.0	$q_{1,1,2}$	0.0	$b_{2,1,1}^{C,out}$	35.73108549568924
$A_{2,2,1}^\beta$	0.0	$q_{1,2,1}$	0.0	$b_{2,2,1}^{C,out}$	0.0
$A_{2,2,2}^\beta$	0.0	$q_{2,1,1}$	0.0	$b_{2,2,2}^{C,out}$	0.0
$A_{hu,2}^\beta$	0.0	$q_{2,2,1}$	0.0		
$A_{1,1,2}$	0.0	$q_{2,2,2}$	0.0		
$A_{1,2,1}$	0.0	$q_{hu,2}$	0.0		
$A_{2,1,1}$	0.0	$t_{1,1,2}^H$	370.0		
$A_{2,2,1}$	0.0	$t_{1,2,1}^H$	650.0		
$A_{2,2,2}$	0.0	$t_{2,1,1}^H$	590.0		
$A_{hu,2}$	0.0	$t_{2,2,1}^H$	590.0		
$dt_{1,1,3}$	240.0	$t_{2,2,2}^H$	466.68393514135914		
$dt_{1,2,1}$	150.0	$t_{1,1,2}^C$	410.0		
$dt_{2,1,1}$	24.166666666666668	$t_{1,2,1}^C$	500.0		
$dt_{2,2,1}$	90.0	$t_{2,1,1}^C$	467.06587160769317		
$dt_{2,2,2}$	220.0	$t_{2,2,1}^C$	350.0		
$dt_{2,2,3}$	240.0	$t_{2,2,2}^C$	440.0112035847521		
$dt_{hu,2}$	180.0	$b_{1,1,2}^{H,in}$	0.0		
$f_{1,1,2}^H$	0.0	$b_{1,2,1}^{H,in}$	4920.152856496002		
$f_{1,2,1}^H$	7.569465933070772	$b_{2,1,1}^{H,in}$	11800.0		
$f_{2,1,1}^H$	20.0	$b_{2,2,1}^{H,in}$	0.0		
$f_{2,2,1}^H$	0.0	$b_{2,2,2}^{H,in}$	0.0		
$f_{2,2,2}^H$	0.0	$b_{1,1,2}^{H,out}$	0.0		
$f_{1,1,2}^C$	4.775206494402774	$b_{1,2,1}^{H,out}$	4920.152856496001		
$f_{1,2,1}^C$	13.0	$b_{2,1,1}^{H,out}$	11800.0		
$f_{2,1,1}^C$	0.06253636270056973	$b_{2,2,1}^{H,out}$	0.0		